

SmartUmrah : Project Management Case Study

Prepared by: Hala Abdullatif Alghamdi
Computer Science Student Princess Nourah University

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1. Introduction:

Every year, millions of Muslims visit Makkah to perform Umrah, which creates significant challenges in managing crowd movement and organizing visitor access to the holy sites [2]. To address these challenges, digital platforms have been introduced to help organize permits and improve the overall experience for visitors. One of the main platforms currently used is the Nusuk application, which allows users to book Umrah permits and access different services related to their visit [1]. The application helps regulate entry and organize appointments; however, there are still opportunities to improve how users plan their visits and how crowd conditions are managed. Therefore, this project proposes an enhanced system called SmartUmrah, which aims to improve the Umrah experience by introducing features such as crowd prediction, improved booking guidance, and intelligent notifications.

2. Problem Statement:

The Nusuk system helps organize Umrah visits by allowing users to reserve permits and schedule their appointments [1]. This contributes to controlling the number of visitors and improving the organization of entry to the holy sites.

However, several limitations still affect the efficiency of the system. First, the current system does not provide crowd level predictions, which means users cannot know how crowded a specific time may be before making a reservation. As a result, many visitors may choose similar time slots, leading to increased congestion.

Second, the system provides limited guidance during the booking process, as users are not informed about less crowded times that may offer a better visiting experience. In addition, the application lacks intelligent notifications that could inform users about crowd conditions or important updates related to their reservations.

To address these limitations, the proposed system SmartUmrah introduces several improvements, including crowd prediction, improved booking guidance, intelligent notifications, and alerts. These features aim to help distribute visitors more effectively, assist users in choosing better visiting times, and enhance safety and overall experience during Umrah [2].

3. Methodology:

3.1 Selected SDLC Model:

For this project, the Hybrid Life Cycle using the Incremental Model will be adopted [3]. The hybrid model combines elements of both predictive and adaptive development approaches. In the incremental model, the system is developed and delivered in smaller functional components rather than building the entire system at once.

Each increment goes through the phases of analysis, design, development, and testing before being integrated into the overall system. This approach allows the development team to deliver system features gradually while ensuring that each component works correctly before moving to the next stage.

The proposed system aims to enhance the functionality of the existing application Nusuk by introducing several new features. These features include predicting crowd levels in advance, providing intelligent notifications about crowd conditions, and offering improved booking guidance based on predicted demand. Each of these features can be developed and implemented as separate increments within the system.

3.2 Justification:

The Hybrid Incremental Model was selected because it is well suited for systems that consist of multiple independent features that can be developed and delivered gradually [3]. The proposed project introduces several enhancements to the existing system, including crowd prediction, intelligent notification services, and improved booking guidance. Developing these features incrementally allows the project team to focus on one component at a time and ensure its quality before integrating it into the complete system.

Another reason for selecting this model is that it improves project control and reduces risk. Since the system is built in increments, testing and evaluation can occur after each stage, allowing the team to identify potential issues early in the development process. This approach also makes it easier to refine system features based on stakeholder feedback.

4. Project Charter:

Project charter for Smart Umrah project

Project Title: SmartUmrah: Intelligent Crowd System for Umrah Pilgrims.

Date of Authorization: March 1

Project Start Date: March 1

Projected Finish Date: June 6

Key Schedule Milestones:

- Complete research and analysis of the current Nusuk application by April 4
- Design smart features for crowd prediction and alerts by April 5
- Develop the prototype for the improved system by May 9
- Conduct user testing and collect feedback by May 23
- Deliver the final project report and presentation by June 6

Budget Information: This project will be developed by the student team using existing software tools and resources.
No external funding is required. All development and design activities will be conducted internally.

Project Manager: Hala Alghamdi, (966)558288737, 445008709@pnu.edu.sa

Project Objectives: The objective of this project is to improve the Nusuk application by adding intelligent features that enhance crowd management and visitor safety during Umrah.

Main Project Success Criteria: The project will be considered successful if:

- The proposed system effectively predicts crowd levels
- Users receive timely safety alerts and notifications
- The system improves awareness of crowded areas
- The solution enhances user experience and safety for visitors

Approach:

- Analyze the current Nusuk application and identify its limitations
 - Design smart features to improve crowd management
 - Develop a conceptual system or prototype
 - Evaluate the proposed solution through analysis and feedback
- Present the final project findings and recommendations

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ministry of Hajj and Umrah	Sponsor	Government Authority	info@haj.gov.sa
Hala Alghamdi	Project Manager	Manager	
Confidential	Team Member	UI/UX Designer	
Confidential	Team Member	System Analyst	
Confidential	Team Member	Developer	
Confidential	Team Member	Documentation	

Sign-off: (Signatures of all the above stakeholders)	
<i>Hala Abdullatif</i> [Redacted Signature]	[Redacted Signature]
Comments: (Handwritten or typed comments from above stakeholders, if applicable)	
<i>" SmartUmrah will enhance the Umrah experience by improving crowd management ensuring more organized journey."</i>	

Figure 1: Project charter

5. Project Scope Statement:

PROJECT SCOPE STATEMENT	
Project Title: SmartUmrah	Date Prepared: May 2026
Project Scope Description: The SmartUmrah project aims to enhance the existing Umrah permit and visit management systems by introducing intelligent features that improve crowd management and user experience. The system focuses on assisting pilgrims in planning their visits more efficiently by providing crowd level predictions, smart booking guidance, and real-time notifications. The solution will act as an enhancement layer to current digital platforms (such as Nusuk), helping users make informed decisions about their visit timing while supporting better distribution of crowds across time slots. The system will be developed incrementally, where each feature (crowd prediction, notifications, booking guidance) will be implemented, tested, and integrated step by step.	
Project Deliverables: • Crowd Prediction Module (predicts expected crowd levels for time slots) • Smart Booking Guidance System (recommends less crowded times) • Intelligent Notification System (alerts users about crowd conditions and updates) • User Interface for displaying predictions and recommendations • Integrated system prototype demonstrating all features • Testing and evaluation reports for each increment • Final project documentation	
Product Acceptance Criteria: • The system must accurately display predicted crowd levels for different time slots • Users must receive clear recommendations for less crowded booking options • Notifications must be delivered in a timely and relevant manner • The system interface must be user-friendly and easy to navigate • Each module (prediction, booking guidance, notifications) must function correctly after integration • The system must improve user decision-making compared to the current process	
Project Exclusions: • Full replacement of the Nusuk application • Real-time tracking of individual users inside the Grand Mosque • Integration with external government systems beyond simulation/prototype level • Payment processing or financial transaction features • Physical crowd control mechanisms (hardware solutions)	
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PROJECT SCOPE STATEMENT
Project Constraints • Limited development time within the academic project duration • Availability of real crowd data may be restricted • Technical limitations in implementing highly accurate prediction models • Dependence on simulated or historical data for testing • Team resources and skill levels
Project Assumptions • Users will have access to smartphones and internet connectivity • Crowd data (historical or simulated) is sufficient for building prediction models • Users are willing to follow system recommendations • The system will be used as a support tool alongside existing platforms • Notifications will improve user awareness and decision-making
Page 2 of 2

Figure 2: Scope Statement

6. Techniques for Collecting Requirements with Justification:

To collect the requirements for the SmartUmrah system, several stakeholders were identified, including Umrah pilgrims, Ministry of Hajj and Umrah staff, system administrators, and technical developers. Different requirement investigation techniques were selected to ensure accurate understanding of user needs and system expectations.

The first technique used is Interviews. Interviews allow direct communication with stakeholders to understand their problems, expectations, and suggestions regarding the current Umrah management systems. This technique was selected because it provides detailed information about user experiences, especially regarding crowd management, booking difficulties, and notification services.

The second technique is Questionnaires. Online questionnaires can be distributed to Umrah visitors to collect feedback from a large number of users in a short time. This technique helps identify common issues faced by users and allows the project team to analyze user preferences related to booking times, crowd conditions, and system usability.

Another technique is Observation. Observing how users interact with existing systems such as Nusuk helps identify practical problems that may not be mentioned directly by users. This technique was chosen because it provides a realistic understanding of user behavior during the booking and permit process.

Finally, Document Analysis was used to review existing information related to the Nusuk platform, crowd management procedures, and Umrah services provided by the Ministry of Hajj and Umrah. This technique helps the team understand the current system structure and identify areas that require improvement.

These techniques were selected because they provide both qualitative and quantitative information, which supports accurate requirement collection and improves the overall quality of the proposed SmartUmrah system.

7. Project Requirements:

7.1 Software requirements:

- Operating System: Windows 10 or later.
- Development Tools: Visual Studio Code or Android Studio.
- Programming Languages: Java, Python, and JavaScript.
- Database Management System: MySQL.
- Web Technologies: HTML, CSS, and React.
- API Services: Google Maps API for location and crowd tracking.
- Notification Services: Firebase Cloud Messaging (FCM).
- Project Management Tools: Microsoft Project for planning and scheduling activities.

These software tools are necessary to develop the crowd prediction system, booking guidance features, and intelligent notification services efficiently.

7.2 Hardware requirements:

- Developer computers with at least Intel Core i5 processor.
- Minimum 8 GB RAM.
- Minimum 256 GB storage capacity.
- Stable internet connection.
- Cloud server for storing and processing system data.
- Smartphones for testing the mobile application features and notifications.

These hardware requirements ensure system stability, efficient processing, and smooth application performance.

7.3 Functional and Non-Functional requirements:

Functional Requirements:

The SmartUmrah system should provide the following functions:

- The system shall allow users to book Umrah permits online.
- The system shall predict crowd levels for different time slots.
- The system shall recommend less crowded times for users.
- The system shall send intelligent notifications and alerts to users.
- The system shall display crowd information using maps and visual indicators.
- The system shall allow administrators to manage booking data and monitor crowd conditions.
- The system shall store user reservations and booking history securely.

Non-Functional Requirements:

- The system shall provide high availability and reliability during peak seasons.
- The system shall respond quickly to user requests and display results efficiently.
- The system shall ensure data privacy and secure user information.

The system shall support large numbers of concurrent users.

- The system shall provide a user-friendly and simple interface.
- The system shall be scalable for future improvements and additional services.
- The system shall maintain accurate notification delivery and real-time updates.

8. Work Breakdown Structure (WBS):

- **Tabular:**

1.1. Project Planning (2 weeks)

- 1.1.1. Define project scope and objectives.
- 1.1.2. Identify system requirements.
- 1.1.3. Develop project charter.
- 1.1.4. Prepare project scope statement.

1.2. Research and Analysis (3 weeks)

- 1.2.1. Analyze limitations of the current Nusuk system.
- 1.2.2. Conduct research on crowd management systems.
- 1.2.3. Identify user needs and system expectations.
- 1.2.4. Define functional and non-functional requirements.
- 1.2.5. Determine system features and functionality.

1.3. System Design and Development (5 weeks)

- 1.3.1. Design system architecture.
- 1.3.2. Design user interface and user experience.
- 1.3.3. Create database structure.
- 1.3.4. Develop crowd prediction module.
- 1.3.5. Develop smart booking guidance feature.
- 1.3.6. Develop intelligent notification system.
- 1.3.7. Integrate all system modules.

1.4. System Testing and Evaluation (2 weeks)

- 1.4.1. Test system functionality and performance.
 - 1.4.2. Conduct usability testing.
 - 1.4.3. Identify and fix system errors.
 - 1.4.4. Evaluate system efficiency and reliability.

1.5. Documentation and Final Delivery (2 weeks)

1.5.1. Prepare technical documentation.

1.5.2. Prepare final project report.

1.5.3. Create project presentation slides.

1.5.4. Submit final project deliverables.

• Diagram:

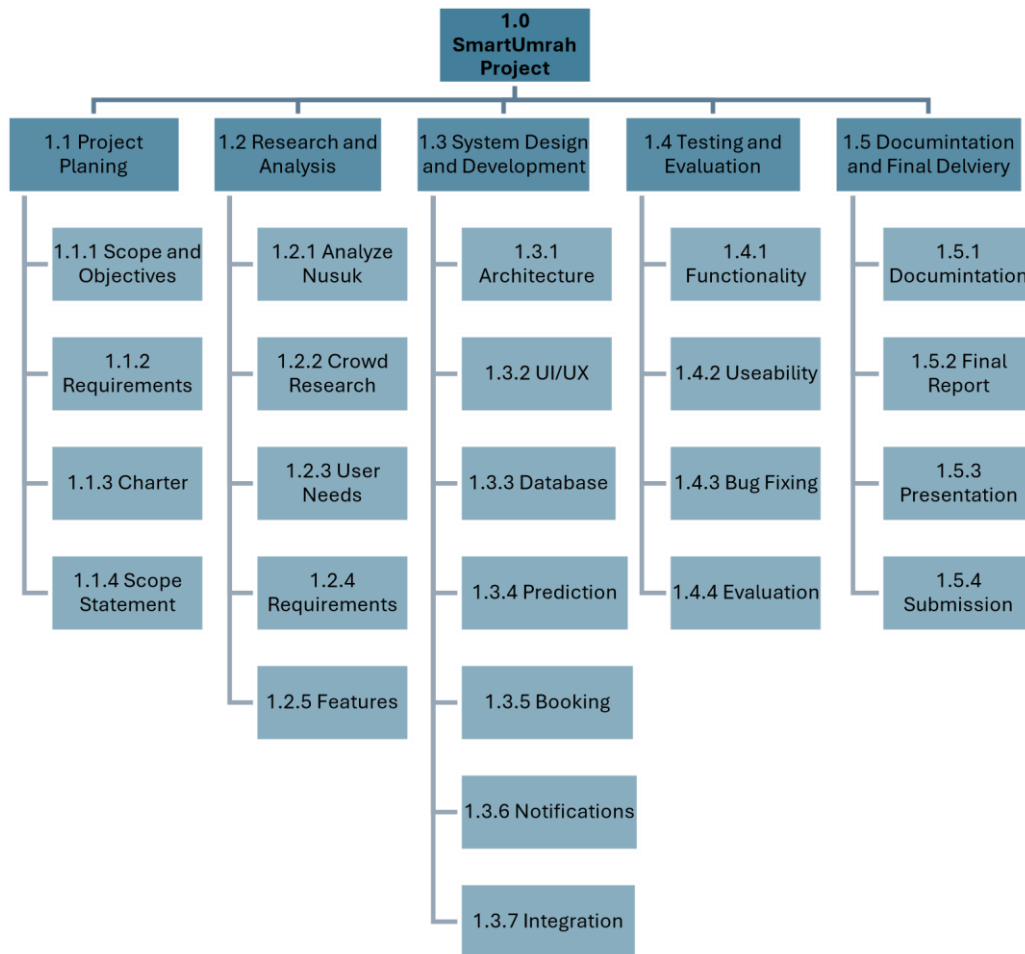


Figure 3: Work Breakdown Structure (WBS) Diagram

9. WBS Dictionary:

WBS ID	Task Name	Description
1.1	Project Planning	Define the overall project objectives, scope, and planning activities.
1.1.1	Define Project Scope and Objectives	Identify the purpose, goals, and expected outcomes of the SmartUmrah project.
1.1.2	Identify System Requirements	Gather system requirements needed for development and implementation.
1.1.3	Develop Project Charter	Prepare the project charter including stakeholders, goals, and responsibilities.
1.1.4	Prepare Project Scope Statement	Define project boundaries, assumptions, and exclusions.
1.2	Research and Analysis	Conduct research and analyze the current Umrah management systems and user needs.
1.2.1	Analyze Limitations of the Current Nusuk System	Study weaknesses and limitations in the existing system.
1.2.2	Conduct Research on Crowd Management Systems	Review existing crowd management and prediction solutions.
1.2.3	Identify User Needs and System Expectations	Determine what users expect from the SmartUmrah system.
1.2.4	Define Functional and NonFunctional Requirements	Specify system functionalities and quality requirements.
1.2.5	Determine System Features and Functionality	Define the major system features and services.
1.3	System Design and Development	Design and develop the SmartUmrah system components and modules.
1.3.1	Design System Architecture	Define system structure and communication between modules.
1.3.2	Design User Interface and User Experience	Create user-friendly screens and navigation flow.

1.3.3	Create Database Structure	Design database tables and relationships.
1.3.4	Develop Crowd Prediction Module	Implement functionality for predicting crowd levels.
1.3.5	Develop Smart Booking Guidance Feature	Implement recommendation system for selecting less crowded times.
1.3.6	Develop Intelligent Notification System	Implement notification and alert services.
1.3.7	Integrate All System Modules	Combine all developed modules into one integrated system.
1.4	System Testing and Evaluation	Test the system and evaluate performance and usability.
1.4.1	Test System Functionality and Performance	Verify that all system functions operate correctly.
1.4.2	Conduct Usability Testing	Evaluate ease of use and user satisfaction.
1.4.3	Identify and Fix System Errors	Detect and correct bugs and technical issues.
1.4.4	Evaluate System Efficiency and Reliability	Measure system stability and reliability.
1.5	Documentation and Final Delivery	Prepare final documentation and submit all deliverables.
1.5.1	Prepare Technical Documentation	Document system design, implementation, and technical details.
1.5.2	Prepare Final Project Report	Complete the academic project report.
1.5.3	Create Project Presentation Slides	Prepare presentation materials for project demonstration.
1.5.4	Submit Final Project Deliverables	Submit all required project files and documents.

Table1 : WBS Dictionary

10. RACI Chart:

WBS Task	Project Manager	UI/UX Designer	System Analyst	Developer	Documentation	Sponsor
1.1. Project Planning						
1.1.1. Define project scope and objectives	R/A	I	C	I	C	I
1.1.2. Identify system requirements	A	C	R	C	I	I
1.1.3. Develop project charter	R/A	I	I	I	C	I
1.1.4. Prepare project scope statement	R/A	I	C	I	C	I
1.2. Research and Analysis						
1.2.1. Analyze limitations of current Nusuk system	A	I	R	I	C	I
1.2.2. Conduct research on crowd management	A	I	R	C	C	I
1.2.3. Identify user needs and expectations	A	R	R	I	I	C
1.2.4. Define functional and non-functional requirements	A	C	R	C	I	I

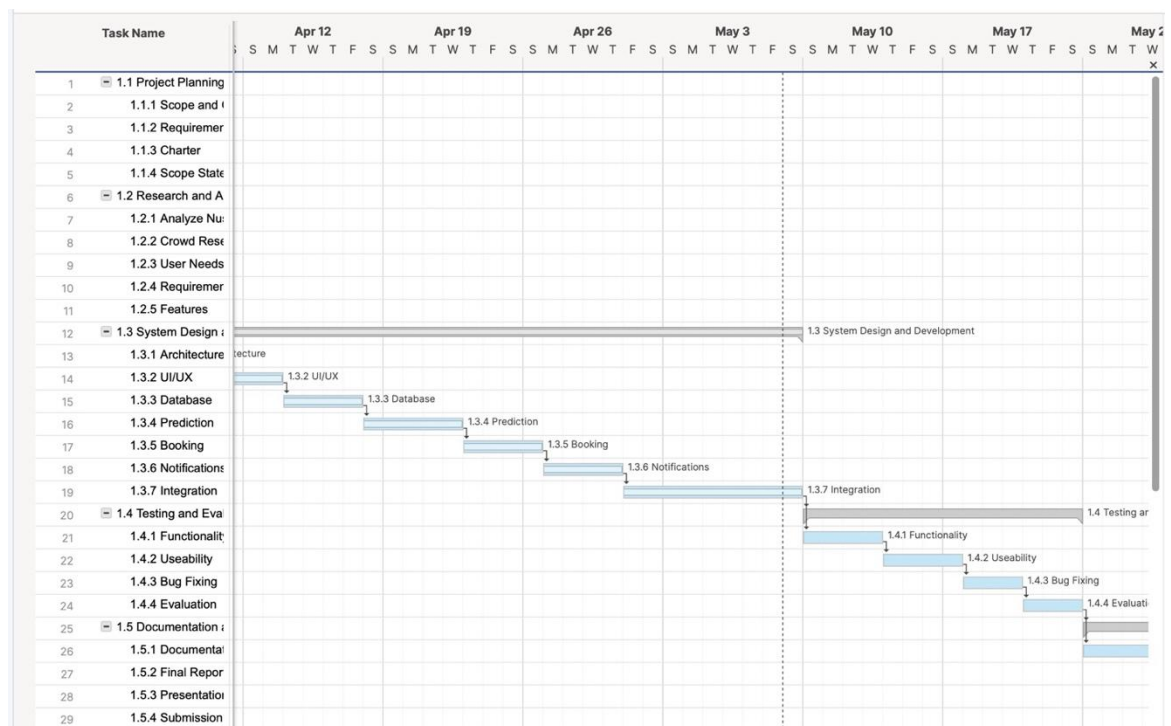
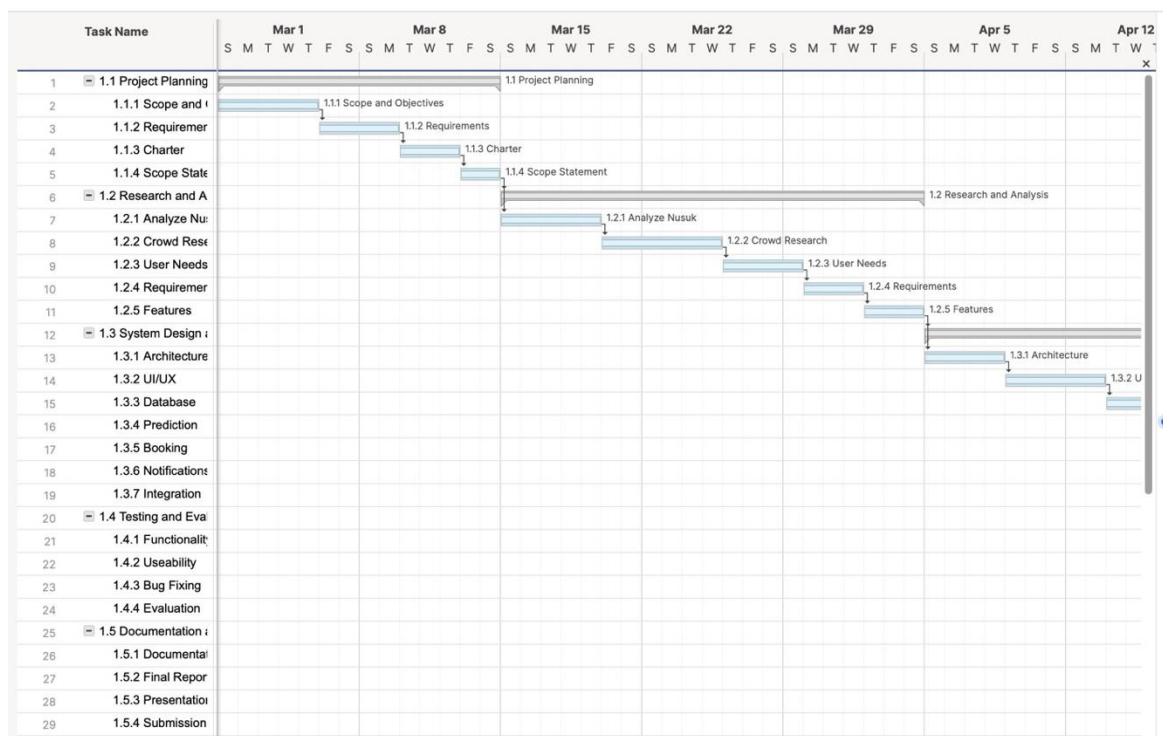
1.2.5. Determine system features and functionality	A	C	R	C	I	C
1.3. System Design and Development						

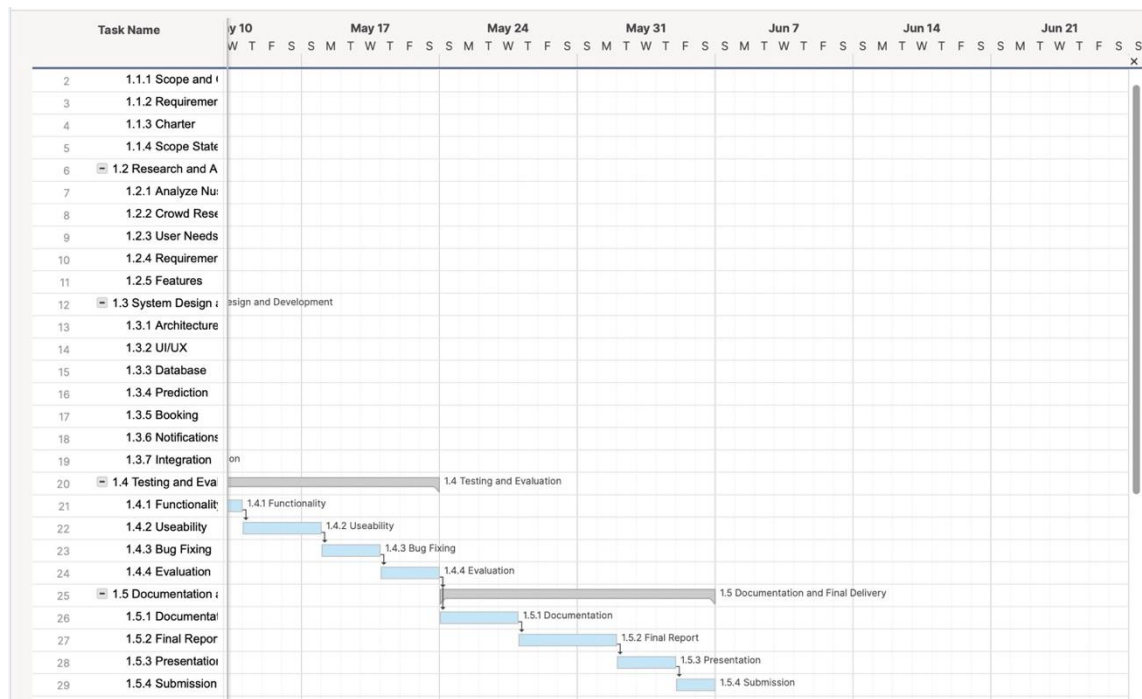
WBS Task	Project Manager	UI/UX Designer	System Analyst	Developer	Documentation	Sponsor
1.3.1. Design system architecture	A	I	R	R	I	I
1.3.2. Design user interface and experience (UI/UX)	A	R	C	I	I	I
1.3.3. Create database structure	A	I	C	R	I	I
1.3.4. Develop crowd prediction module	A	I	I	R	I	I
1.3.5. Develop smart booking guidance feature	A	I	I	R	I	I
1.3.6. Develop intelligent notification system	A	I	I	R	I	I
1.3.7. Integrate all system modules	A	C	C	R	I	I

1.4. System Testing and Evaluation						
1.4.1. Test system functionality and performance	A	I	R	R	I	I
1.4.2. Conduct usability testing	A	R	R	I	I	C
1.4.3. Identify and fix system errors	A	C	C	R	I	I
1.4.4. Evaluate system	A	I	R	I	I	C
WBS Task	Project Manager	UI/UX Designer	System Analyst	Developer	Documentation	Sponsor
efficiency and reliability						
1.5. Documentation and Final Delivery						
1.5.1. Prepare technical documentation	A	I	I	R	R	I
1.5.2. Prepare final project report	A	C	C	C	R	I
1.5.3. Create project presentation slides	R	R	R	R	R	I
1.5.4. Submit final project deliverables	R	I	I	I	I	A

Table 2: RACI Chart

11• Gantt Chart:





Gantt Chart Table:

	Task Name	Duration	Start	Finish	Predecesso...	%Complete	Status
1	1.1 Project Planning	14d	01/03/26	14/03/26		1	Complete
2	1.1.1 Scope and Requirements	5d	01/03/26	05/03/26		100%	Complete
3	1.1.2 Requirement Gathering	4d	06/03/26	09/03/26	2	100%	Complete
4	1.1.3 Charter	3d	10/03/26	12/03/26	3	100%	Complete
5	1.1.4 Scope Statement	2d	13/03/26	14/03/26	4	100%	Complete
6	1.2 Research and Analysis	21d	15/03/26	04/04/26	5	1	Complete
7	1.2.1 Analyze Needs	5d	15/03/26	19/03/26	5	100%	Complete
8	1.2.2 Crowd Research	6d	20/03/26	25/03/26	7	100%	Complete
9	1.2.3 User Needs	4d	26/03/26	29/03/26	8	100%	Complete
10	1.2.4 Requirement Gathering	3d	30/03/26	01/04/26	9	100%	Complete
11	1.2.5 Features	3d	02/04/26	04/04/26	10	100%	Complete
12	1.3 System Design	35d	05/04/26	09/05/26	11	1	Complete
13	1.3.1 Architecture	4d	05/04/26	08/04/26	11	100%	Complete
14	1.3.2 UI/UX	5d	09/04/26	13/04/26	13	100%	Complete
15	1.3.3 Database	4d	14/04/26	17/04/26	14	100%	Complete
16	1.3.4 Prediction	5d	18/04/26	22/04/26	15	100%	Complete
17	1.3.5 Booking	4d	23/04/26	26/04/26	16	100%	Complete
18	1.3.6 Notifications	4d	27/04/26	30/04/26	17	100%	Complete
19	1.3.7 Integration	9d	01/05/26	09/05/26	18	100%	Complete
20	1.4 Testing and Evaluation	14d	10/05/26	23/05/26	19	0	Not Started
21	1.4.1 Functionality	4d	10/05/26	13/05/26	19	0	Not Started
22	1.4.2 Useability	4d	14/05/26	17/05/26	21	0	Not Started
23	1.4.3 Bug Fixing	3d	18/05/26	20/05/26	22	0	Not Started
24	1.4.4 Evaluation	3d	21/05/26	23/05/26	23	0	Not Started
25	1.5 Documentation and Final Delivery	14d	24/05/26	06/06/26	24	0	Not Started
26	1.5.1 Documentation	4d	24/05/26	27/05/26	24	0	Not Started
27	1.5.2 Final Report	5d	28/05/26	01/06/26	26	0	Not Started
28	1.5.3 Presentation	3d	02/06/26	04/06/26	27	0	Not Started
29	1.5.4 Submission	2d	05/06/26	06/06/26	28	0	Not Started

Table 3 :Gantt chart

12.Risk Register:

No .	Ran k	Risk	Description	Categor y	Root Cause	Triggers	Potential Responses	Probabili ty	Impact if the Risk Occurs
R1	1	Delay in crowd updates	Delay in updating crowd data may affect the accuracy of information shown to users	Negativ e	Server overload or slow data processing	Slow response time or delayed data synchronization	Improve data update speed and use reliable servers	H	H
R2	2	System overload during Umrah seasons	High traffic on the system may cause slow performance or temporary service interruption	Negativ e	Large number of concurrent users	Sudden increase in active users during peak times	Use backup cloud servers and monitor system performance	M	H
R3	3	Future system expansion	The system may successfully expand by adding more smart services in the future	Positive	System scalability and successful performance	Increased demand for additional smart services	Plan for future scalability and feature expansion	M	M
R4	4	Increase d user adoption	Users may increasingly rely on the system because of ease of use	Positive	Simple interface and accurate notifications	Positive feedback and increasing number of users	Continuously improve user experience and notification services	H	M

Table 4 : Risk register

13. References:

- [1] Ministry of Hajj and Umrah, "Nusuk Platform," <https://www.nusuk.sa>
- [2] Ministry of Hajj and Umrah, "Hajj and Umrah Services," <https://www.haj.gov.sa>
- [3] K. Schwalbe, "Information Technology Project Management," <https://accord.edu.so/web/content/33460>